int

int64_t

SharedPtr

SharedPtr

vector< smacc2::ISmaccState * >

shared_ptr< smacc2 ::introspection::SmaccState Info >

SmaccStatus

map< std::string, std ::shared_ptr< smacc2 ::ISmaccOrthogonal > >

shared_ptr< smacc2 ::introspection::SmaccState MachineInfo >

recursive_mutex

atomic< StateMachineInternal Action >

map< void *, std::shared _ptr< smacc2::CallbackCounterSemaphore > >

map< std::string, std ::pair< std::function < std::string()>, boost ::any > >

vector< smacc2_msgs ::msg::SmaccTransitionLogEntry >

SMRunMode

SharedPtr

-reactorWalkDepth_

-stateSeqCounter_

#nh_

#timer_

#currentState_

#currentStateInfo_

#status_msg_

#orthogonals_

#stateMachineInfo_

-eventQueueMutex_ -m_mutex_

-stateMachineCurrentAction

-stateCallbackConnections

-globalData_

-transitionLogHistory_

-runMode_

#eventsLogPub_ #stateMachinePub _ #stateMachineStatus Pub_ #transitionHistory Service_ #transitionLogPub_

smacc2::ISmaccStateMachine

+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ joinAsyncClientBehaviorThreads()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfiguration Finished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()

mutex

vector< std::shared _ptr< smacc2::ISmaccClient > >

vector< std::vector < std::shared_ptr< smacc2 ::ISmaccClientBehavior > > >

double

atomic< bool >

-statusPub_

vector< smacc2::ISmaccUpdatable * >

vector< std::vector < smacc2::ISmaccUpdatable * > >

SmaccFifoScheduler *

processor_handle

thread

ExecutionModel

-mutex_

#clients_

-clientBehaviors_

-stateMachine_

-smaccStateMachine_

-signalDetector_

-loop_rate_hz

-end_ -initialized_ -rosInitialized_

-updatableClients_

-updatableStateElements_

-scheduler_

-processorHandle_

-signalDetectorThread_

-executionModel_

smacc2::ISmaccOrthogonal

+ ~ISmaccOrthogonal()
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()
- currentClientBehaviors()

smacc2::SignalDetector

+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ joinAsyncClientBehaviorThreads()
+ pollingLoop()
+ postEvent()
+ queueWorkItem()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyRosInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()

boost::signals2::signal < Signature, optional _last_value< typename boost ::function_traits< Signature >::result_type >, int, std ::less< int >, function< Signature >, typename extended_signature< function_traits< Signature >::arity, Signature >::function_type, boost:: signals2::mutex >

map< smacc2::ComponentKey, std::shared_ptr< smacc2 ::ISmaccComponent > >

-stateMachine_

-orthogonal_

#components_

#stateMachine_

smacc2::ISmaccClient

+ ISmaccClient()
+ ~ISmaccClient()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
# onComponentInitialization()
# onStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()

+clearClouds +minHits +triggerClouds

boost::signals2::signal < void(), optional_last _value< typename boost:: function_traits< void()> ::result_type >, int, std ::less< int >, function< void ()>, typename extended_signature < function_traits< void()>::arity, void()>::function_type, boost::signals2::mutex >

boost::signals2::signal < void(const MessageType &), optional_last_value < typename boost::function _traits< void(const MessageType &)>::result_type >, int, std ::less< int >, function< void(const MessageType &)>, typename extended _signature< function_traits< void(const MessageType &)>::arity, void(const MessageType &)>::function_type, boost::signals2::mutex >

smacc2::SmaccSignal < Signature, Combiner, Group, GroupCompare, SlotFunction, ExtendedSlotFunction, Mutex >

boost::signals2::signal < void(const sensor_msgs ::msg::PointCloud2 &), optional _last_value< typename boost:: function_traits< void(const sensor _msgs::msg::PointCloud2 &)>::result _type >, int, std::less< int >, function < void(const sensor_msgs::msg::PointCloud2 &)>, typename extended_signature< function _traits< void(const sensor_msgs::msg::PointCloud2 &)>::arity, void(const sensor_msgs::msg::PointCloud2 &)>::function_type, boost::signals2::mutex >

-sub_

-clearStreak_ -gatedClouds_ -triggerStreak_

< void()>

< void(const MessageType &)>

#owner_

< void(const sensor _msgs::msg::PointCloud2 &)>

smacc2::ISmaccComponent

+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()

-attitudeSub_

-attitudeMutex_

smacc2::SmaccSignal < void()>

bool

function< void(const MessageType &)>

SharedPtr

smacc2::SmaccSignal < void(const MessageType &)>

string

optional< int >

smacc2::SmaccSignal < void(const sensor _msgs::msg::PointCloud2 &)>

-lastYaw_

-tooClose_

+clearRangeM +floorCutoffM +halfAngleRad +maxYawRateRadS +minRangeM +triggerRangeM

+levelFrame +opticalFrameInput

-firstMessage_ -initialized_

+postInitialMessageEvent +postMessageEvent

-sub_

+onFirstMessageReceived

+onMessageReceived_

-topicName_

+queueSize

smacc2::client_core _components::CpTopicSubscriber < MessageType >

+ CpTopicSubscriber()
+ CpTopicSubscriber()
+ ~CpTopicSubscriber()
+ onMessageReceived()
+ onFirstMessageReceived()
+ onStateOrthogonalAllocation()
+ onInitialize()
- messageCallback()

function< void(const sensor_msgs::msg::PointCloud2 &)>

cl_px4_mr::ForwardObstacle GuardParams

-cosHalfAngle_

+onObstacleCleared

+onObstacleTooClose_

-haveAttitude_

< sensor_msgs::msg ::PointCloud2 >

+postInitialMessageEvent +postMessageEvent

smacc2::client_core _components::CpTopicSubscriber < sensor_msgs::msg::PointCloud2 >

+ CpTopicSubscriber()
+ CpTopicSubscriber()
+ ~CpTopicSubscriber()
+ onMessageReceived()
+ onFirstMessageReceived()
+ onStateOrthogonalAllocation()
+ onInitialize()
- messageCallback()

-params_

-subscriber_

uint64_t

array< float, 4 >

atomic< float >

atomic< int >

-lastYawStampUs_

-attitudeQ_

-lastMinRange_ -yawRate_

-lastHits_

cl_px4_mr::CpForwardObstacle Guard

+ CpForwardObstacleGuard()
+ ~CpForwardObstacleGuard()
+ onInitialize()
+ onObstacleTooClose()
+ onObstacleCleared()
+ reset()
+ isTooClose()
+ lastYawRate()
+ lastMinRange()
+ lastHits()
+ params()
- onCloud()
- onAttitude()
- levelRotation()